

LEDGER · 2026-08-31 · BIOTECHNOLOGY

Enzymatic Cyclic Phosphatidic Acid (cPA) via Biocatalytic Transphosphatidylation

REJECTED

This concept did **not** clear the framework.
The reasoning is published in full below.
What the assessment found

Incumbent benchmark	Bulk Retinol (Vitamin A)
Entry application	Premium anti-aging cosmetic active
Measured advantage	Reduces TEWL by 13-15% over 14 days compared to Retinol which increases TEWL, but fails price-parity requirement as it requires a \$350/kg price point.
Time to first revenue	12 months
Gross margin at parity	-18.5%
Startup CapEx	\$270,000
Capital productivity	2.64x
Regulatory risk	Moderate — qualification costs reported (\$150,000 estimated)
Why it failed	

- Voided by: Economics undisclosed, Negative control / sub-2x
- Margin fails the operating-cost check

Assessment

The proposal fails unit margins and startup CapEx criteria. The raw LPC feedstock costs alone (\$45/kg) result in an OPEX of \$82.94/kg, producing negative gross margins at Retinol's market price of \$70/kg. Startup CapEx (\$270,000) also exceeds the \$250,000 threshold.

Also flagged

- Duplicate references inflating the evidence base
- Price parity asserted, not sourced

A rejection is not a claim that the science is wrong. It means the venture case did not survive: the comparator was not what the buyer actually buys, the comparison was not like-for-like, the advantage fell short of a step change, or the capital and time to first revenue were prohibitive.

Assessed 2026-08-31 against seven gates plus the voiding sub-tests. [How the method works](#) · [Browse all concepts](#)

The Absence Audit

Method

Ledger

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Track record

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